1. Exercise

Linear models and regression

Exercise Sheet 1

Institute of Mathematics
Prof. Dr. Matthias Schulte
M.Sc. Florent Onana Assouga

Winter Term 2025/26

October 16, 2025

Solve the following problems before the exercise class on October 23.

Problem 1
For $n \in \mathbb{N}$ let $T_n \sim \chi^2_n$.

a) Compute $\mathbf{E}[T_n]$ and $\mathbf{Var}[T_n]$.
Hint: You may use that $\mathbf{E}[N^4] = 3$ *for a standard normal random variable* N.

b) Prove that

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{T_n - n}{\sqrt{n}} \leq t\right) = \Phi(t/\sqrt{2})$$

for $t \in \mathbb{R}$, where Φ is the distribution function of the standard normal distribution.

Problem 2
Let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$ such that $\mathbf{E}[\|X\|^2] < \infty$. Prove that the covariance matrix $\mathbf{Cov}(\mathbf{X})$ of $\mathbf{X}$ is positive semi-definite, i.e.,

$$\mathbf{a}^T \mathbf{Cov}(\mathbf{X})\mathbf{a} \geq 0$$

for all $\mathbf{a} \in \mathbb{R}^m$.

Problem 3
Let $\mathbf{X} = (X_1, X_2)^T \sim \text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} \quad \text{and} \quad \boldsymbol{\Sigma} = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}.$$

Compute $\mathbf{P}(2X_1 + 2X_2 \leq 2, X_1 - X_2 \leq 5)$.

Problem 4
Consider a possibly unfair coin with the probability $p \in (0,1)$ for heads and the probability $1-p$ for tails. The coin is tossed n times independently. Let H_n and T_n denote the numbers of heads and tails, respectively, and define $\hat{p}_n = \frac{H_n}{n}$.

a) Compute $\mathbf{E}[\hat{p}_n]$.

b) Show that

$$\hat{p}_n \xrightarrow{\mathbf{P}} p \quad \text{as} \quad n \to \infty \quad \text{and} \quad \hat{p}_n \xrightarrow{\text{a.s.}} p \quad \text{as} \quad n \to \infty.$$

Problem 1

$n \in \mathbb{N} \qquad T_n \sim \chi^2_n$

a) We recall that $T_n = \sum_{i=1}^{n} N_i^2$ where

$N_i \quad i = 1, \dots, n$ are i.i.d r.v. $\sim \mathcal{N}(0,1)$

$$E[T_n] = E\left[\sum_{i=1}^{n} N_i^2\right]$$
$$= \sum_{i=1}^{n} E[N_i^2]$$
$$= \sum_{i=1}^{n} 1 = n$$

remember variance of i.i.d $\mathcal{N}(0,1)$

and

$$Var[T_n] = Var\left[\sum_{i=1}^{n} N_i^2\right]$$
$$= \sum_{i=1}^{n} Var[N_i^2] \quad \text{[since independent]}$$
$$= \sum_{i=1}^{n} \left(E[N_i^4] - E[N_i^2]^2\right)$$

$E[N_i^2] \to 1$

$E[N_i^4] \to 3$ - based on hint

$$= \sum_{i=1}^{n} (3-1)$$
$$= \sum_{i=1}^{n} 2$$
$$= 2n$$

$$\boxed{\therefore \; E[T_n] = n \quad \& \quad Var[T_n] = 2n}$$

$\frac{1}{n}$ ... $\lim$ ... $P(S_n - nE[X] < t)$

$S_n = \sum_{i=1}^{n} N_i \leftarrow \lim_{n\to\infty} \mathbb{P}\left(\frac{\ldots}{\sqrt{Var[X]}/n} \ldots\right) = \Phi(t)$

b) $\rightarrow$ Recall central limit theorem

Since N_i $i=1,\ldots n$ are iid, from central limit theorem we have

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{T_n - n}{\sqrt{2n}} \leq t'\right) = \Phi(t')$$

for every $t' \in \mathbb{R}$

$$\Leftrightarrow \quad \lim_{n\to\infty} \mathbb{P}\left(\frac{T_n - n}{\sqrt{n}} \leq \sqrt{2}\, t'\right) = \Phi(t')$$

for $t' = \frac{t}{\sqrt{2}}$

we have

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{T_n - n}{\sqrt{n}} \leq \sqrt{2} \times \frac{t}{\sqrt{2}}\right) = \Phi\left(\frac{t}{\sqrt{2}}\right)$$

Problem 2

$x \in \mathbb{R}^m$

Let $a = (a_1, \ldots a_m)^T \in \mathbb{R}^m$

We have

$$a^T Cov(x) = (a_1, \ldots a_m) \begin{pmatrix} Cov(x_1, x_1) & \ldots & Cov(x_1, x_m) \\ \vdots & \ddots & \vdots \end{pmatrix}$$

$$\begin{pmatrix} Cov(x_m, x_1) & \dots & Cov(x_m, x_m) \end{pmatrix}$$

$$= \left(\sum_{i=1}^{m} a_i Cov(x_i, x_1), \dots, \sum_{i=1}^{m} a_i Cov(x_i, x_m) \right)$$

Therefore

$$a^T Cov(x) a = \left(\sum_{i=1}^{m} a_i Cov(x_i, x_1) \dots \sum_{i=1}^{m} a_i \cdot Cov(x_i, x_m) \right) \begin{pmatrix} a_1 \\ \vdots \\ a_m \end{pmatrix}$$

$$= a_1 \sum_{i=1}^{m} a_i Cov(x_i, x_1) + \dots + a_m \sum_{i=1}^{m} a_i Cov(x_i, x_m)$$

$$= \sum_{i=1}^{m} \sum_{j=1}^{m} a_j a_i (x_i, x_j)$$

Recall

$$Cov(aX+b, cY+d) = ac\, Cov(X, Y)$$

$$= Cov\left(\sum_{i=1}^{m} a_i x_i, \sum_{j=1}^{m} a_j x_j \right)$$

$$= Var\left[\sum_{i=1}^{m} a_i x_i \right] \geq 0$$

Variance is non-negative

$$\boxed{\therefore \quad a^T Cov(x) a \geq 0}$$

Note: vector is always a column matrix

$\therefore$ we had $a = (a_1, \dots a_n)^T \in R^m$

Problem 3

We have $x = (x_1, x_2)^T \quad x \sim \mathcal{N}(\mu, \Sigma)$

$$\mu = (3, -2)^T \quad \Sigma = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$$

we set $Y = (Y_1, Y_2)^T$ such that

$$Y_1 = 2X_1 + 2X_2 \qquad Y_2 = X_1 - X_2$$

we have $Y = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix} X$

i.e., $\Rightarrow Y = AX$

Since $x \sim \mathcal{N}(\mu, \Sigma)$

$$Y = AX \sim \mathcal{N}(A\mu, A\Sigma A^T)$$

$$A\mu = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$$A\Sigma = \begin{pmatrix} 2 & 2 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} 16 & 16 \\ 2 & -2 \end{pmatrix}$$

$$A\Sigma A^T = \begin{pmatrix} 16 & 16 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 64 & 0 \\ 0 & 4 \end{pmatrix}$$

$$Y \sim N\left(\begin{pmatrix} 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 64 & 0 \\ 0 & 4 \end{pmatrix}\right)$$

from lemma 2.2, we can conclude that Y_1 and Y_2 are independent

$$\Rightarrow Y_1 \sim N(2, 64) \quad Y_2 \sim N(5, 4)$$

$$\mathbb{P}(2X_1 + 2X_2 \leq 2, \; X_1 - X_2 \leq 5)$$

$$= \mathbb{P}(Y_1 \leq 2, \; Y_2 \leq 5)$$

$$= \mathbb{P}(Y_1 \leq 2) \cdot \mathbb{P}(Y_2 \leq 5)$$

$$= \frac{1}{2} \cdot \frac{1}{2}$$

$$= \frac{1}{4}$$

Problem 4

$i = 1, \dots n$

we denote by x_i the r.v.

$$x_i = \begin{cases} 1 & \text{with prob } p \\ 0 & \text{with prob } 1-p \end{cases}$$

$x_i \sim \text{Bern}(p)$ are i.i.d

Bernoulli

H_n = number of heads

$$H_n = \sum_{i=1}^{n} x_i$$

$$T_n = n - H_n$$

$H_n \sim \text{Binom}(n, p)$

a) we have

$$E[\hat{P}_n] = E\left[\frac{H_n}{n}\right] = \frac{E[H_n]}{n} = \frac{np}{n}$$

$$= \underline{\underline{p}}$$

b) We have $\hat{P}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$

with x_i $i = 1, \dots n$ i.i.d

From WLLN (Weak law of large numbers)

[Weak Law of Large numbers]

$$\hat{p}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} E[X_i] = p$$

and from the SLLN

[Strong Law of Large numbers]

$$\hat{p}_n \xrightarrow{a.s} p$$

c) a $\Rightarrow$ $\hat{p}_n$ is an unbiased estimator for p

b $\Rightarrow$ $\hat{p}_n$ is strongly and weakly consistent.

d) We have

$$\frac{1}{\sqrt{n}}\begin{pmatrix} H_n - np \\ T_n - n(1-p) \end{pmatrix} = \frac{1}{\sqrt{n}}\left\{\begin{pmatrix} H_n \\ T_n \end{pmatrix} - \begin{pmatrix} np \\ n(1-p) \end{pmatrix}\right\}$$

$$= \frac{1}{\sqrt{n}}\left\{\begin{pmatrix} \sum_{i=1}^{n} X_i \\ n - \sum_{i=1}^{n} X_i \end{pmatrix} - \begin{pmatrix} np \\ n(1-p) \end{pmatrix}\right\}$$

$$= \underbrace{\begin{pmatrix} \sum_{i=1}^{n} \overbrace{\begin{matrix} X_i \\ 1 - X_i \end{matrix}}^{= Y_i} \end{pmatrix} - \overbrace{\begin{pmatrix} np \\ n(1-p) \end{pmatrix}}^{= E[Y]}}$$

$\sqrt{n}$

$$= \frac{\sum_{i=1}^{n} Y_i - E\left[\sum_{i=1}^{n} Y_i\right]}{\sqrt{n}}$$

$Y_1 = \begin{pmatrix} Y_1 \\ 1-Y_1 \end{pmatrix}$ are iid

$$\xrightarrow[\text{where}]{d} \mathcal{N}(0, \Sigma)$$

$$\Sigma = \begin{pmatrix} Var[X_1] & Cov[X_1, 1-X_1] \\ Cov[1-X_1, X_1] & Var[1-X_1] \end{pmatrix}$$

$$= \begin{pmatrix} p(1-p) & -p(1-p) \\ -p(1-p) & p(1-p) \end{pmatrix}$$

$\det \Sigma = |\Sigma| = 0$

and Σ is not invertible